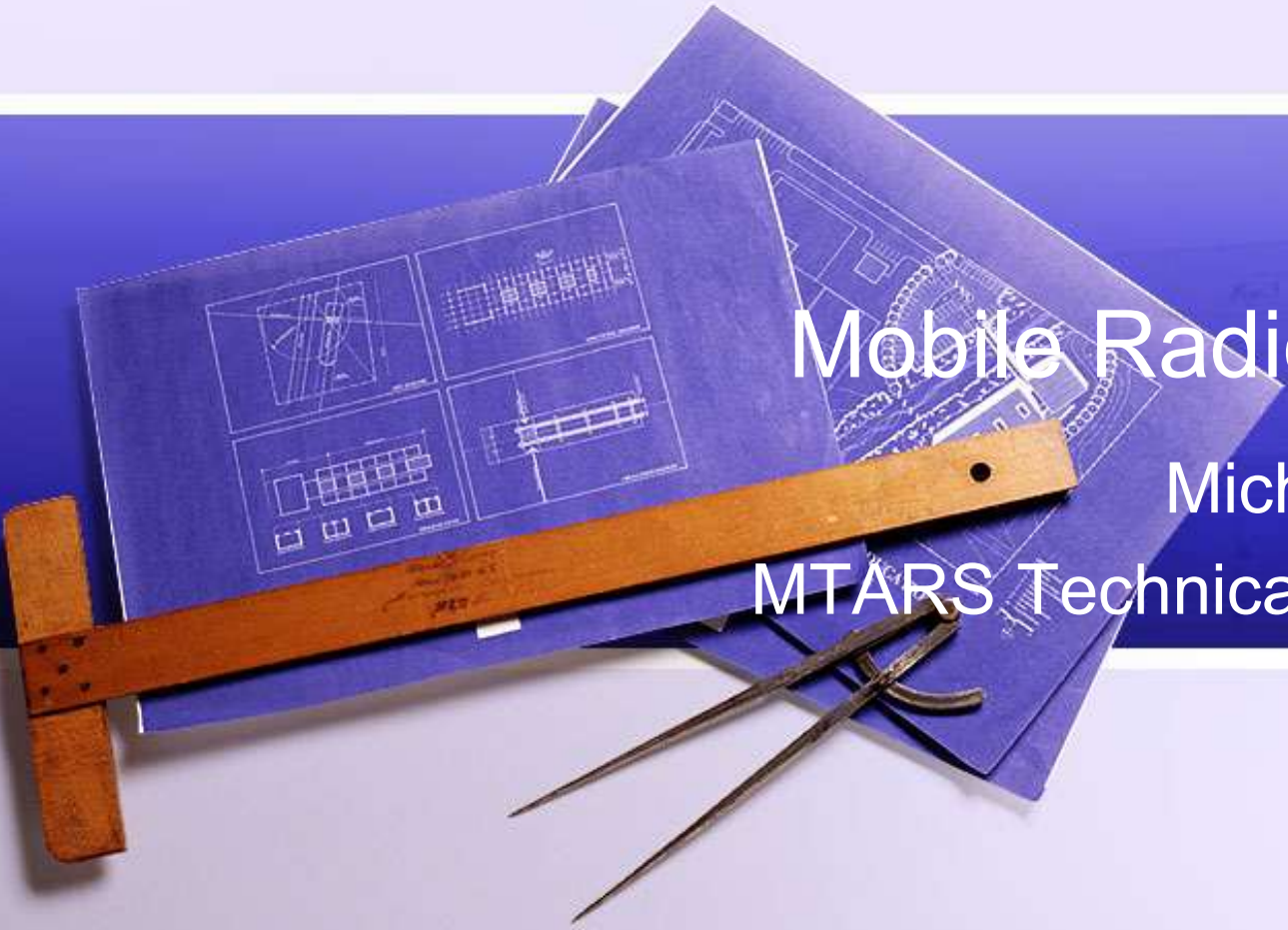


A photograph of technical drawing tools including a wooden T-square, a pair of compasses, and a French curve resting on a set of blueprints. The blueprints contain various technical diagrams, including a perspective view of a building and several cross-sectional or plan views. The background is a solid blue gradient.

Mobile Radio Install Basics

Michael Hamby KI4UJY

MTARS Technical Topic Spring 2026

Contents

- Disclaimer - Discussion Limited to VHF/UHF and basics for HF (limited to simple antenna/Power)
- Operation Safety
- Purpose — Old school ham's just assume every new ham can figure out mobile radio mounting — this is not the case
- 12 Volt Power Distribution/Routing
- Antenna Location and mounting VHF/UHF for the purpose of brevity
- Radio Mounting
- Discussion
- References

Mobile Radio Safety

- Operator Distraction –
 - **Preprogrammed Radio – use mic to go up and down channels**
 - **Recommend staying on one frequency**
 - Especially true for HF
 - Pull over to adjust – Glance at radio don't stare!
 - **Nice to have remote heads at dash height**
 - May be impractical
 - **Don't Operate your vehicle distracted especially in severe weather!**
 - **Use may be challenged by law enforcement if car driven erratically**



Cited Law: (written to discourage cell phone distraction)

- **2024 Tennessee Code**
Title 55 - MOTOR AND OTHER
VEHICLES (§§ 55-1-101 — 55-53-105)
Chapter 8 - OPERATION OF VEHICLES
- RULES OF THE ROAD (§§ 55-8-101 —
55-8-307)
Part 1 - OPERATION OF VEHICLES -
RULES OF THE ROAD (§§ 55-8-101 —
55-8-199)
Section 55-8-199 - Prohibited uses of
wireless telecommunications devices
or stand-alone electronic devices

A blue folder with technical drawings and a wooden ruler is visible on the left side of the slide. The folder contains several sheets of paper with technical drawings, including a cross-section diagram and a grid pattern. A wooden ruler is placed over the drawings.

Ham Radio Exemption – by definition

- Wireless telecommunications device” does not include a radio, citizens band radio, citizens band radio hybrid, commercial two-way radio communication device or its functional equivalent, subscription-based emergency communication device, prescribed medical device, amateur or ham radio device, or in-vehicle security, navigation, autonomous technology, or remote diagnostics system.

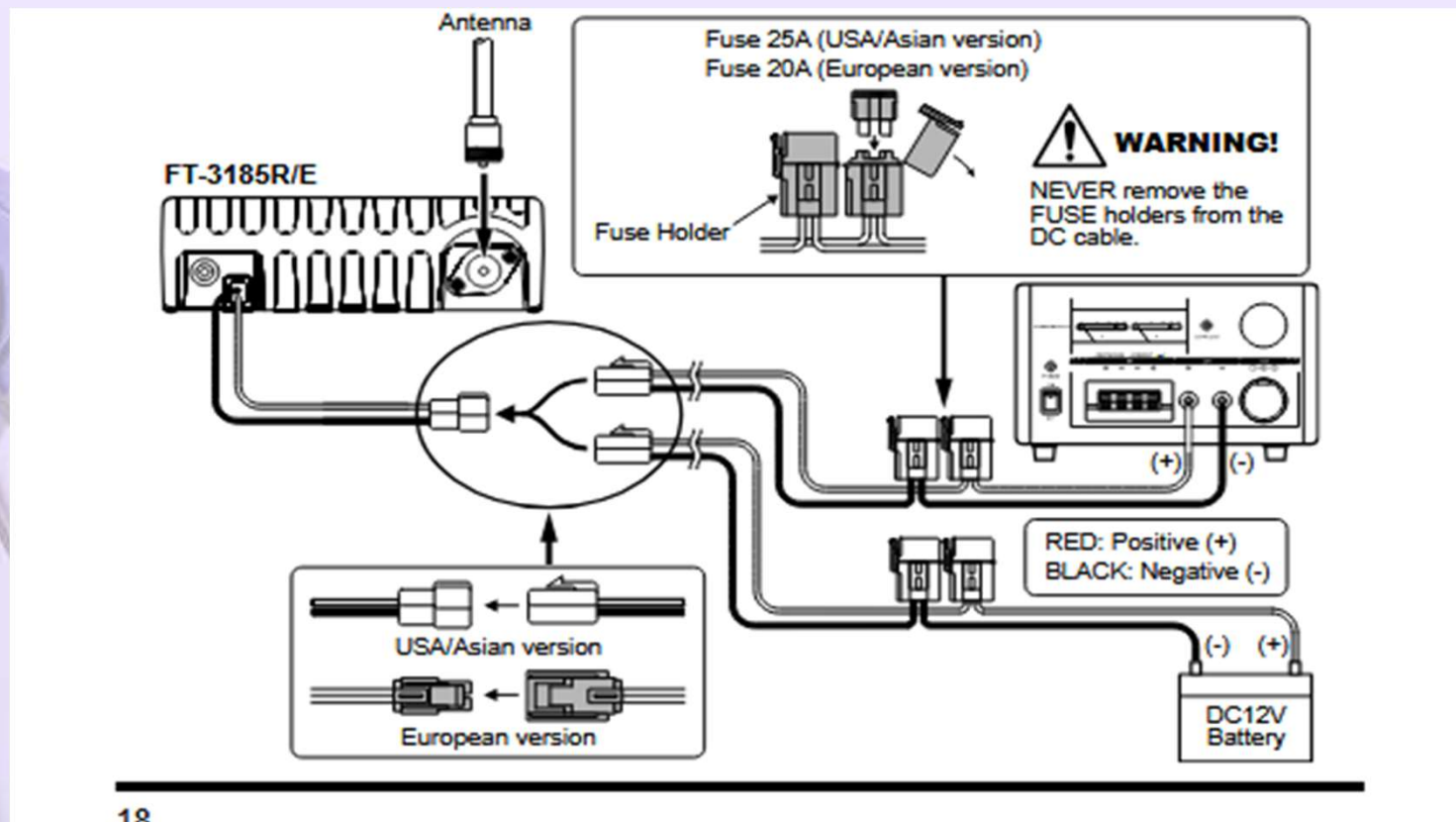
Installation Manufacturers Recommendations

- Read the Radio Manual for installation Instructions!
 - **Direct to Battery(1)**
 - **Power Supply(1)**
 - **Some HT's have a lighter plug adapter**

Note: (1) It is common practice to power your equipment from a power distribution System (DIY or Assembled)

Power Connections

- So Why not the lighter plug?



Cautions

- Do not use a DC power supply cable other than the one that is provided.
- Do not use the DC power supply cable with the fuse holder cut off.
- Use an external power source capable of supplying DC 13.8V, a current capacity of 20A or more.

Distribution — Blocks, Terminals, Buss

- Lots of choices and cost ranges



Installation Types

- Quick and Dirty (first setup for me)
 - HT Operation – Mobile
 - Mag Mount Antenna
- VHF/UHF
 - Simple Radio/ Remote Head Type
 - Mag mount or fixed antenna
- HF
 - Remote Head/One Piece
 - Tuner
 - Antenna
- Adapter/Speaker Mounts

Considerations

- Car Style (car, SUV, Truck, Van)
- Security
- Cost
- Fit and Finish
- Removability
- Safety (operation distraction)
- Power routing from engine compartment
- Antenna routing
- Use trim tools to pop trim up without damage.



Removable Head/Installation

- Remote Head is the most flexible
- Radio should be secured



Radio Bracket Mounting

- Requires something to screw too –
- Personal Preference is remote mount in some type of box or plate under seat
- Old days – sheet metal screws into dash, not recommended today
- Avoid drilling into vital bits of the car (air bags, freon lines, heater core lines electrical)
- Avoid interfering with air bags on dash
- Consider a severe crash



Truck Configuration

- Lumber/hinges –gravity mount
 - **Formerly held FT-987 HF/All mode radio**
 - **Advantage Wood screw mounting**
 - **Modular**
- Current configuration two Radios
 - **VHF for FM Voice**
 - **VHF/UHF for APRS (later model Yeasu)**
- Power by Battery Go Box (50 amp hour)
- Solar recharged (with Lipo4FE controller)
 - **Car Power Lead straight to battery (10 awg) – actually recharged by solar as well via load connection.**
- Dual Band Lip mount on rear door – coax routed through trim
- VHF high gain rear of truck – coax wire tied to frame
- Commet CA-2x4SR Mobile VHF/UHF Antenna and UHF style Mount
- Android Car Sterio with APRSDroid, weather radar etc. Hands free via mic mounted on steering column - DIY

SUV Mount

- Base – plywood/black denim
 - Mounted between seat and center console
 - Ram Mount Hardware – Radio Head - adjustable
 - Remote Head Dual Band Radio – APRS/Voice (FM/Digital Yeasu C4FM)
 - Radio bulk mounted under seat – Amo case
 - Auxiliary Battery –20 amp/hour power box and Anderson Power Pole Distribution (solar recharge)
 - West Mountain Car Battery Bridge (ineffective with Lipo4FE – but maintains at a minimum voltage)
 - Trunk Lip Mount Dual Band Mobile Antenna
 - After Market Extension Cable for Mic
 - After Market remote speaker
 - Road and Wind Noise – (due to cargo assembly on roof)
 - Phone mount/charger
 - APRSDroid, radar app
 - Hands free via Ford SYNC



YL's car rig - temporary

- HT with Mag Mount antenna and cable adapter
- Speaker/Mic
- Rechargeable Battery
- 12 volt pack to replenish or Cig lighter



HF Remote head Yaesu FT-891

- Example of a remote head mount HF radio
- YSK-891 Front Panel Separation Kit



HF antenna – brief review

- Lots of options – dx engineering search
- smallest?
 - Yaesu ATAS-120A Auto Active Tuning Antennas ATAS-120^a
- Manual
 - [Moonraker MD-3500 Sliding Manually Tunable HF/Plus 6m Portable Vertical Antennas 12-124](#)
- Grounding!



Radio Mounting

- Lido LM-AD1 Display Head Adapters LM-AD1
- Ram Mount – Cup holder etc (\$\$)



Discussion

- Special thanks to Michael Glennon – KB4JHU who on several occasion helped me install radio's and taught me the basics.



References

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